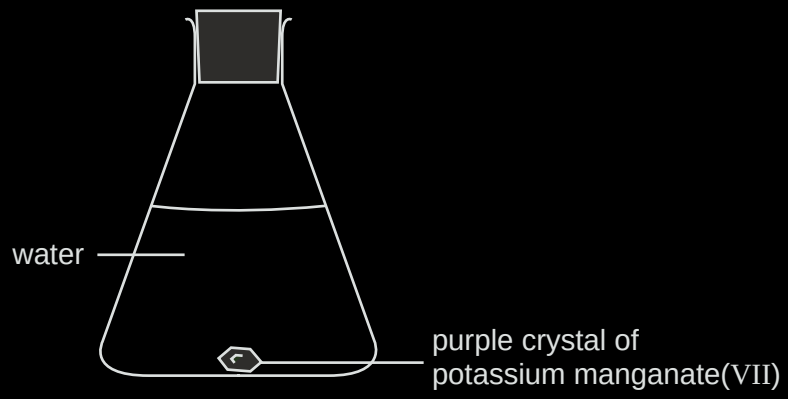


What is observed?

A a colourless layer below a purple layer

B a colourless liquid with the purple crystal unchanged

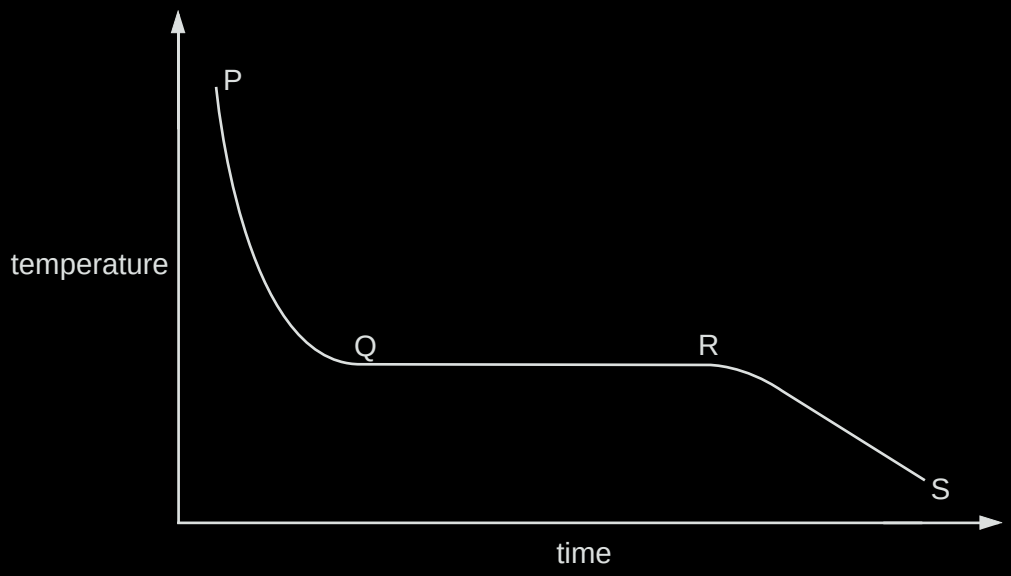
C a purple layer below a colourless layer

D a uniformly purple solution

2 A student adds aqueous sodium hydroxide or aqueous ammonia to aqueous solutions of four different metal compounds.

Which solution contains Zn^{2+} ions?

solution	add a few drops of NaOH(aq)	add excess NaOH(aq)	add a few drops of NH_3(aq)	add excess NH_3(aq)
A	ppt	ppt dissolves	ppt	ppt dissolves
B	ppt	ppt dissolves	ppt	ppt
C	ppt	ppt	no ppt	no ppt
D	no ppt	no ppt	no ppt	no ppt



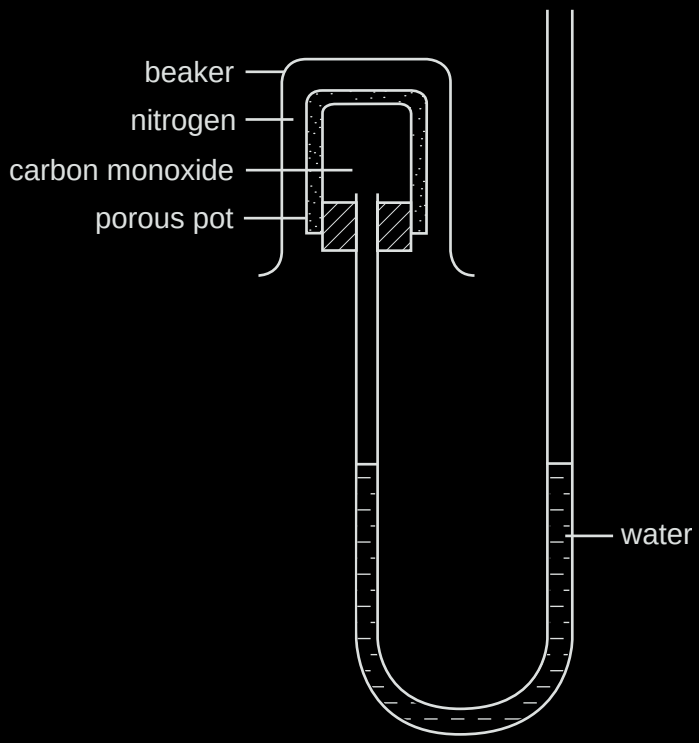
When are liquid and solid both present?

A P to Q and R to S

B P to Q

C Q to R

D R to S



What is the reason for this?

A Both gases are diatomic.

B Nitrogen is an unreactive gas.

C The gas particles are too large to pass through the porous pot.

D The two gases have the same relative molecular mass.

5 Which statement about all the noble gases is correct?

A The number of protons in the atoms equals the number of neutrons.

B Their atoms each have a stable arrangement of electrons.

C Their atoms each have eight electrons in their outer shell.

D They exist as molecules containing two atoms.

6 A substance **Q** conducts electricity both when solid and molten.

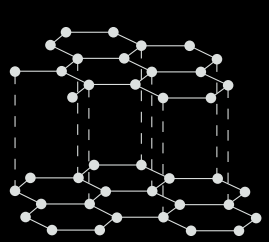
What is **Q**?

A an alloy

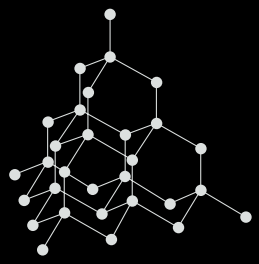
B a hydrocarbon

C a metal oxide

D a salt



S



T

Which set of data is correct for these two structures?

	conducts electricity	very hard material	can be used as lubricant
A	T	T	S
B	S	T	S
C	S	S	T
D	T	S	T

8 Substance **X** has a melting point higher than 500 °C. It is insoluble both in water and in organic solvents. It conducts electricity when both solid and molten.

What is **X**?

A copper

B mercury

C poly(ethene)

D sodium chloride

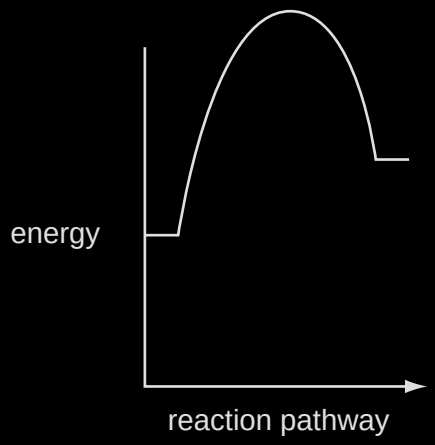
9 How many moles per dm^3 of gaseous carbon dioxide are there if 4.4 g occupies 500 cm^3?

A 0.1 mol / dm^3 **B** 0.2 mol / dm^3 **C** 2.2 mol / dm^3 **D** 8.8 mol / dm^3

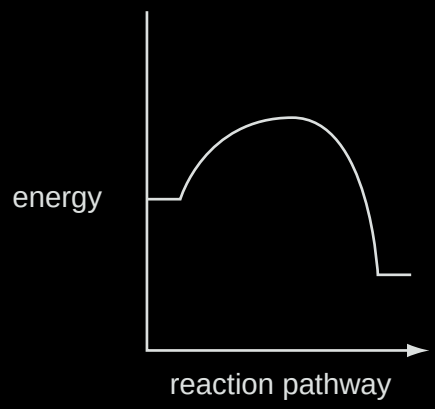
10 Which reactions take place during the electrolysis of aqueous copper(II) sulphate with copper electrodes?

	reaction at positive electrode	reaction at negative electrode
A	$Cu^{2+} + 2e^- \rightarrow Cu$	$Cu \rightarrow Cu^{2+} + 2e^-$
B	$4OH^- \rightarrow 2H_2O + O_2 + 4e^-$	$Cu^{2+} + 2e^- \rightarrow Cu$
C	$Cu \rightarrow Cu^{2+} + 2e^-$	$2H^+ + 2e^- \rightarrow H_2$
D	$Cu \rightarrow Cu^{2+} + 2e^-$	$Cu^{2+} + 2e^- \rightarrow Cu$

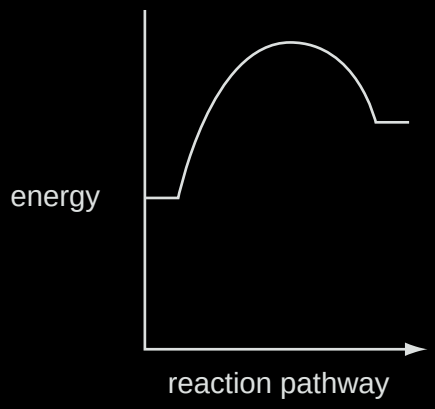
A


energy
reaction pathway

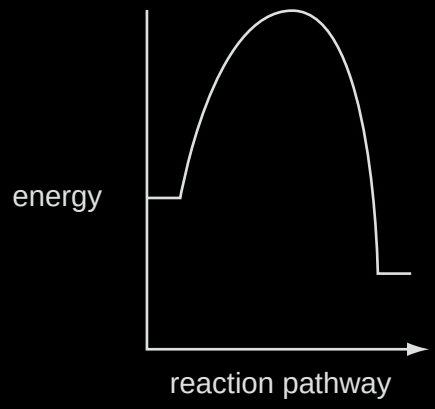
B

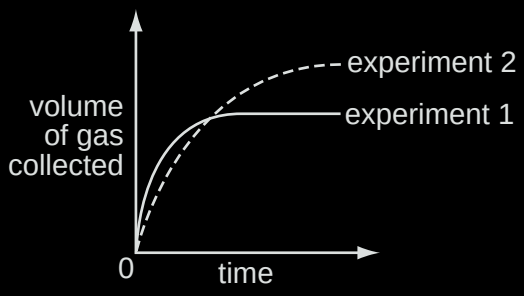

energy
reaction pathway

C

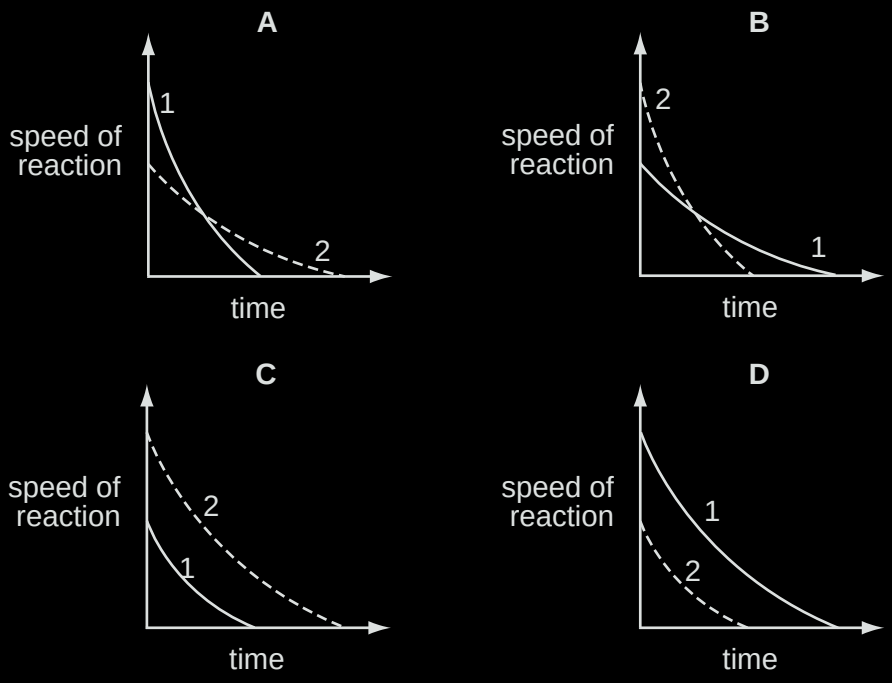

energy
reaction pathway

D


energy
reaction pathway



Which graph shows how the speed of reaction varies with time in each experiment?



15 A colourless gas is passed into each of three different solutions. The results are shown in the table.

solution of	potassium iodide	acidified potassium dichromate(VI)	acidified potassium manganate(VII)
result	stays colourless	orange to green	purple to colourless

What is the colourless gas?

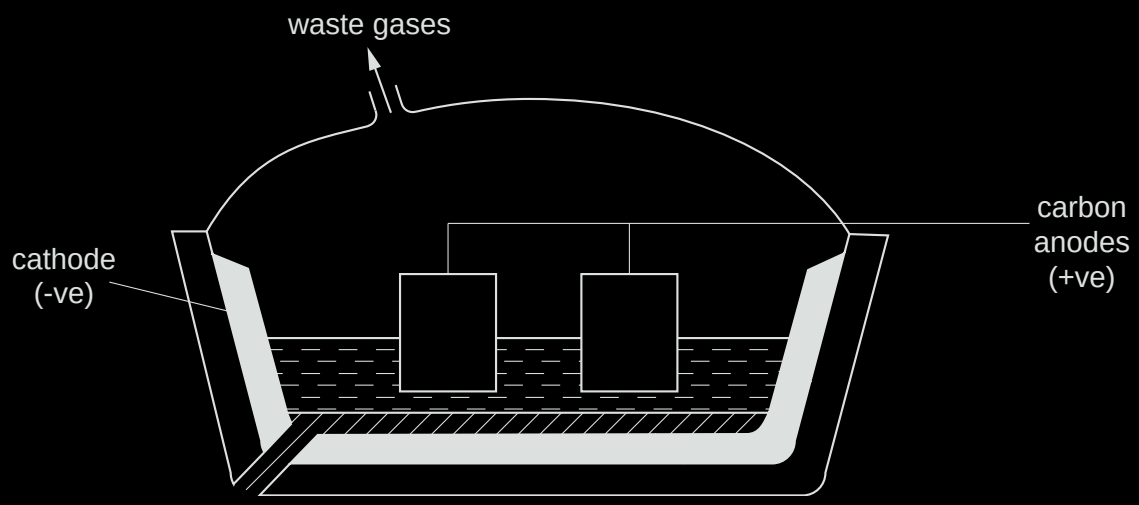
A an acid

B an alkali

C an oxidising agent

D a reducing agent





What is the physical state of the aluminium oxide and aluminium during this process?

	aluminium oxide	aluminium
A	liquid	liquid
B	liquid	solid
C	solid	liquid
D	solid	solid

23 Aluminium is used to make saucepans because of its apparent lack of reactivity.

Which property of aluminium explains its unreactivity?

A It has a high electrical conductivity.

B It has a low density.

C It has a surface layer of oxide.

D It is in Group III of the Periodic Table.

24 Alloys are usually harder than the metals from which they are made.

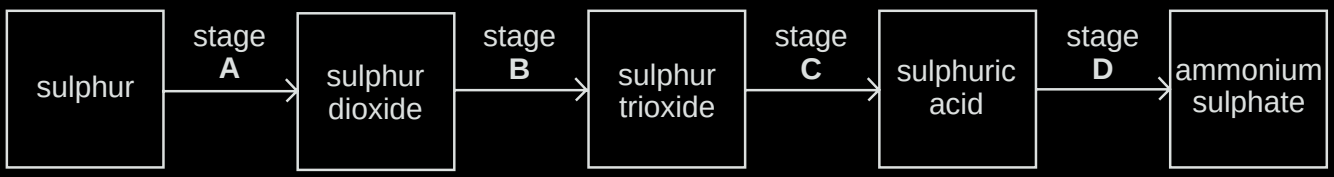
Which difference between the metals explains the greater hardness of alloys?

A atomic radii

B densities

C electrical conductivities

D relative atomic masses



30 The diagram shows the colours of the indicators, methyl orange and methyl red, at different pH values.

pH	2	3	4	5	6
colour of methyl orange	red		yellow		
colour of methyl red	red				yellow

The table shows the pH of four solutions.

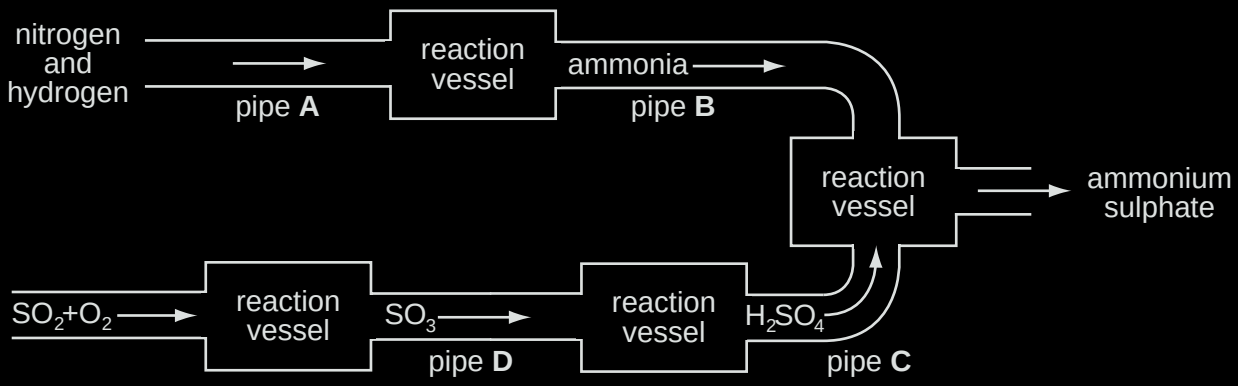
solution	W	X	Y	Z
pH	2	3	5	6

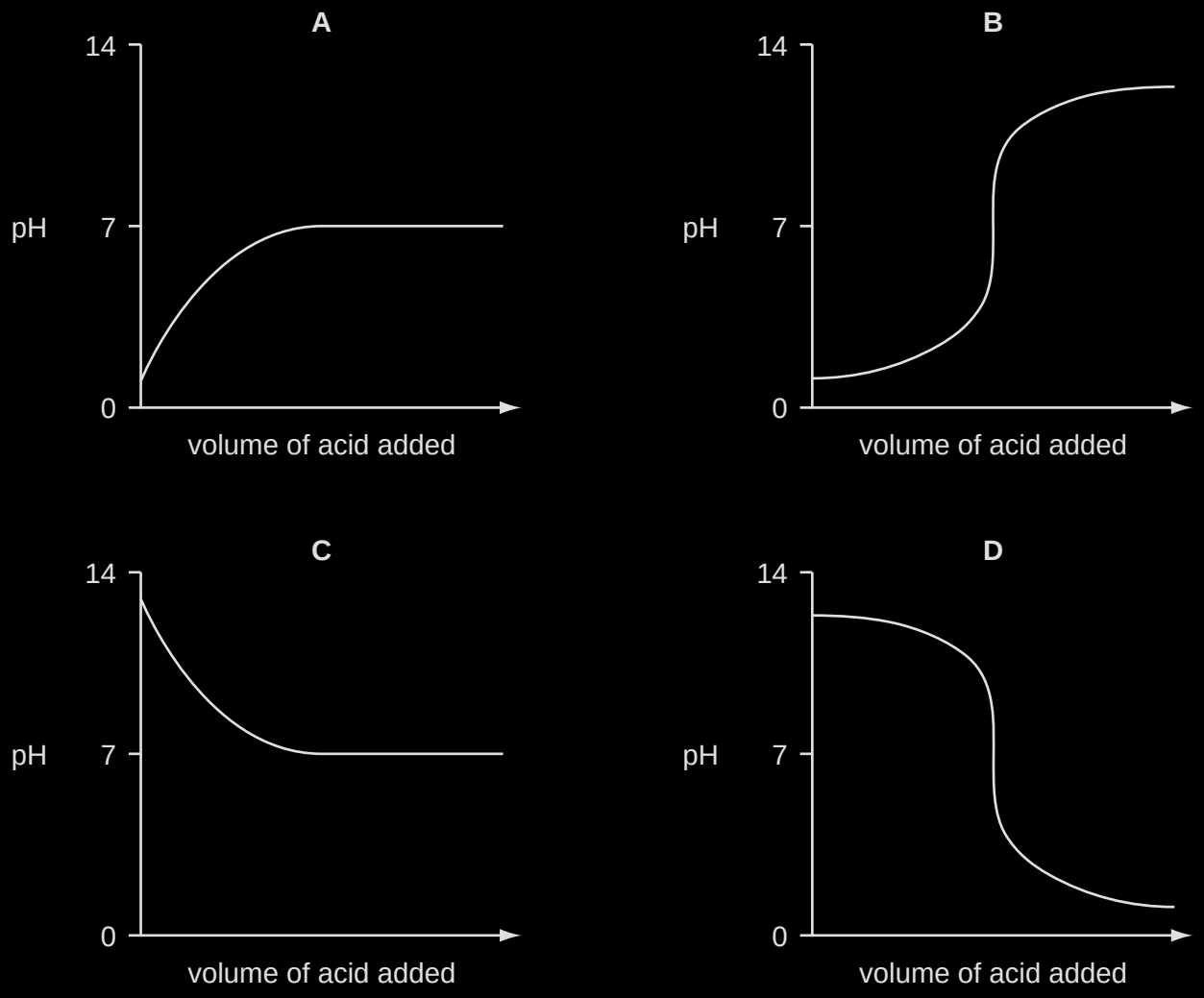
In which solutions will both indicators be yellow?

A W and X **B** X and Y **C** Y and Z **D** Z only

31 The diagram shows some of the stages in the manufacture of ammonium sulphate.

From which connecting pipe would a major leak most **increase** the pH value of rain?





33 Two tests are carried out on a solution of compound **X**.

test	result
add nitric acid followed by aqueous silver nitrate	white precipitate formed
excess aqueous sodium hydroxide added	white precipitate formed that does not re-dissolve

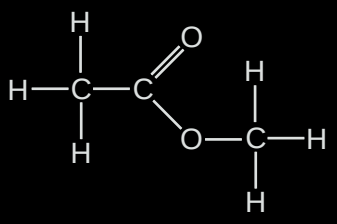
What is compound **X**?

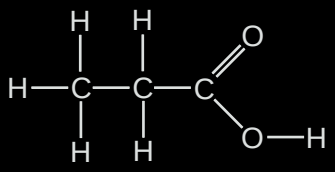
A aluminium chloride

B aluminium sulphate

C calcium chloride

D calcium sulphate

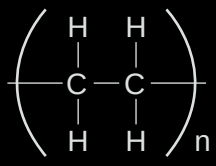




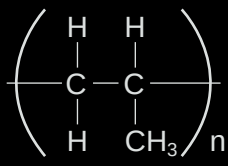
What are the similarities and differences between the two compounds?

	similarities	differences
A	molecular formulae	reactions
B	molecular formulae	relative molecular masses
C	structures	molecular formulae
D	structures	relative molecular masses

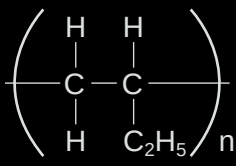
A



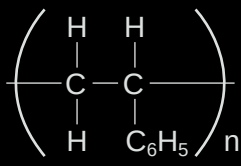
B



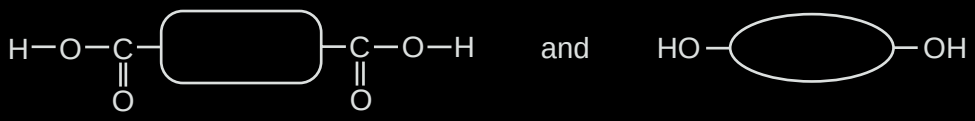
C



D

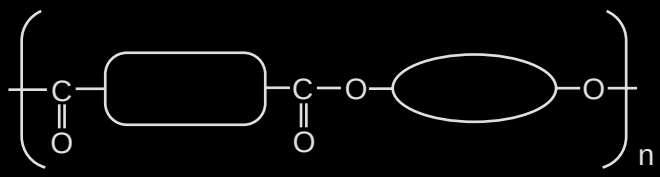


40 *Terylene* (a polyester) is made by condensation polymerisation of the two monomers shown.

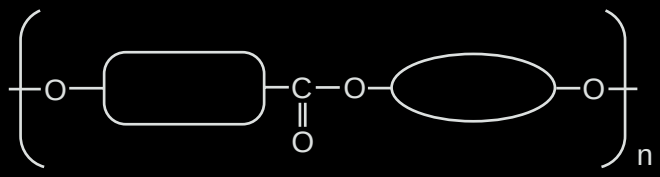


What is the repeat unit of the polymer?

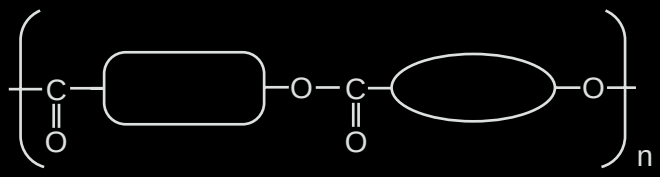
A



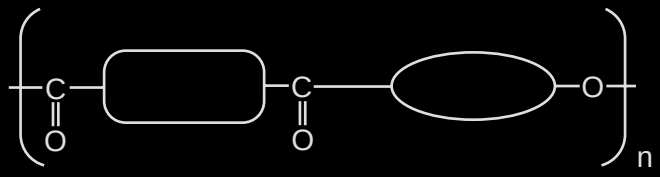
B



C



D



DATA SHEET
The Periodic Table of the Elements

Group																	
I	II											III	IV	V	VI	VII	0
							1 H Hydrogen 1										4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulphur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	64 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
85 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57 *	178 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	Po Polonium 84	At Astatine 85	Rn Radon 86
Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

140 Ce Cerium 58	141 Pr Praseodymium 59	144 Nd Neodymium 60	Pm Promethium 61	150 Sm Samarium 62	152 Eu Europium 63	157 Gd Gadolinium 64	159 Tb Terbium 65	162 Dy Dysprosium 66	165 Ho Holmium 67	167 Er Erbium 68	169 Tm Thulium 69	173 Yb Ytterbium 70	175 Lu Lutetium 71
232 Th Thorium 90	Pa Protactinium 91	238 U Uranium 92	Np Neptunium 93	Pu Plutonium 94	Am Americium 95	Cm Curium 96	Bk Berkelium 97	Cf Californium 98	Es Einsteinium 99	Fm Fermium 100	Md Mendelevium 101	No Nobelium 102	Lr Lawrencium 103

*58-71 Lanthanoid series
90-103 Actinoid series

Key

a
X
b

a = relative atomic mass
X = atomic symbol
b = proton (atomic) number

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).